

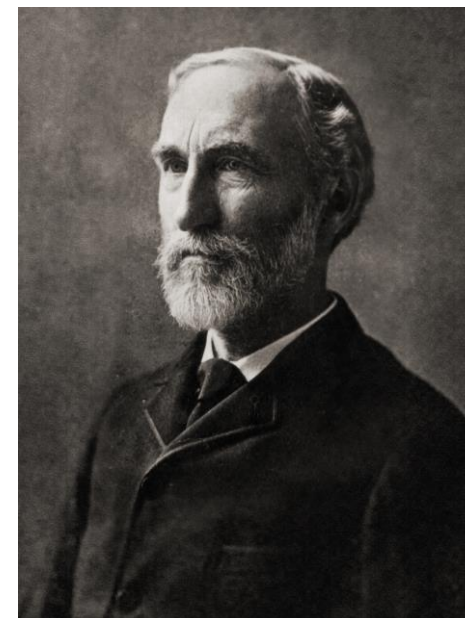
Math Primer

(8. Vectors)

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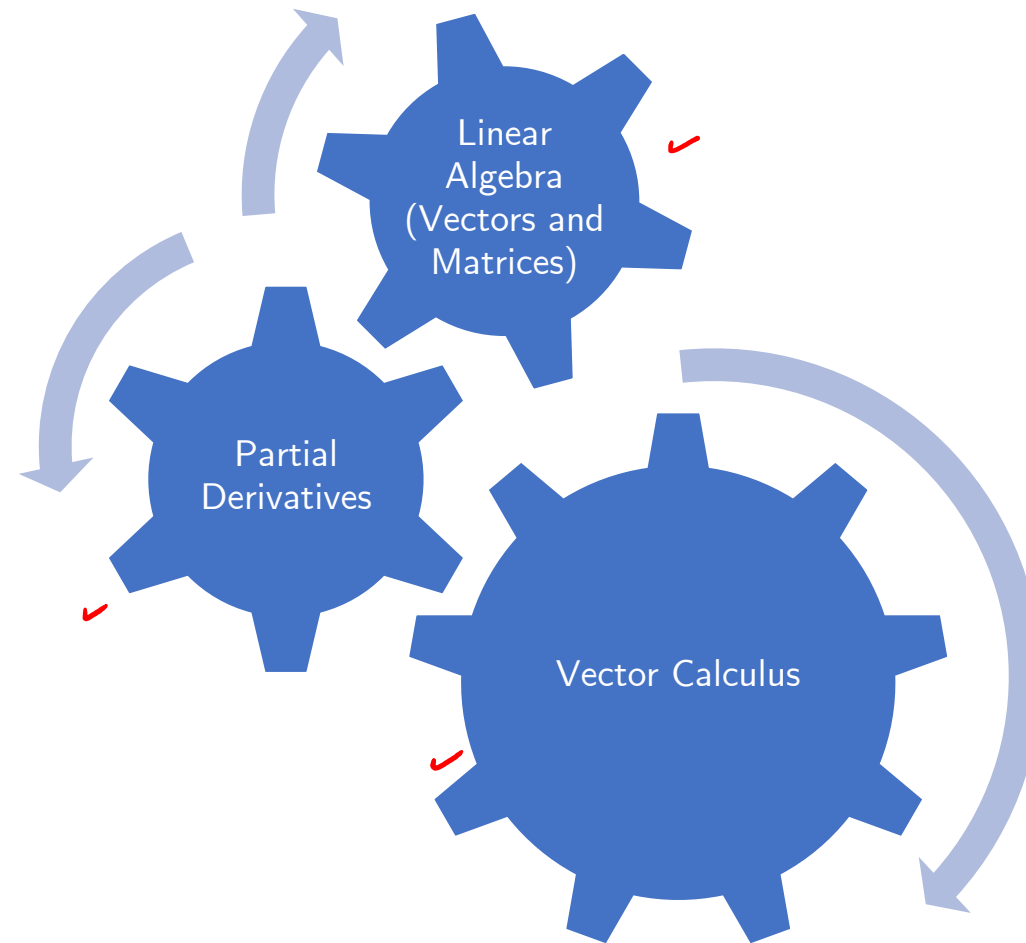


- ▶ Vector representation
- ▶ Vector Addition and Scaling
- ▶ Scalar Product
- ▶ Projection
- ▶ Span
- ▶ Linear Independence
- ▶ Vector Norms



Josiah Willard Gibbs
(1839 – 1903)

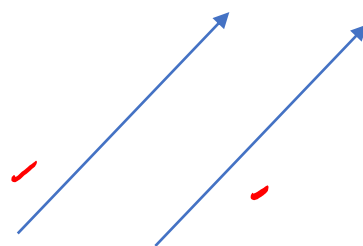
The Multivariable Calculus Landscape



Vectors



- ▶ A Vector is an entity that has **magnitude** and **direction**. Geometrically, one can draw it as an arrow. Length of the arrow is the vector's magnitude, and the direction of the arrow is the vector's direction.



(same vector)

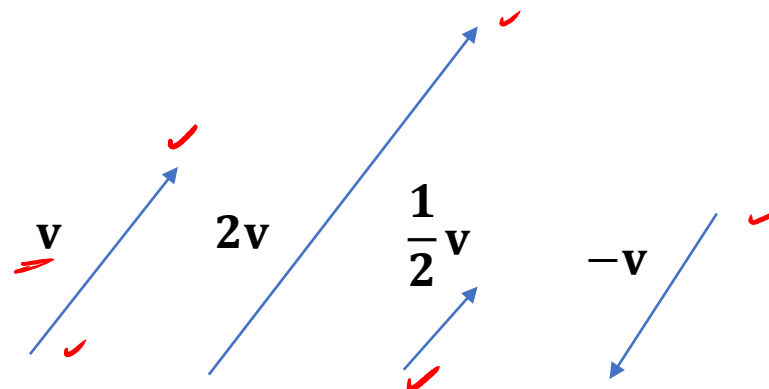
- ▶ We will denote vectors in small bold letters e.g., **v**, and the length of the vector is given as $|\mathbf{v}|$



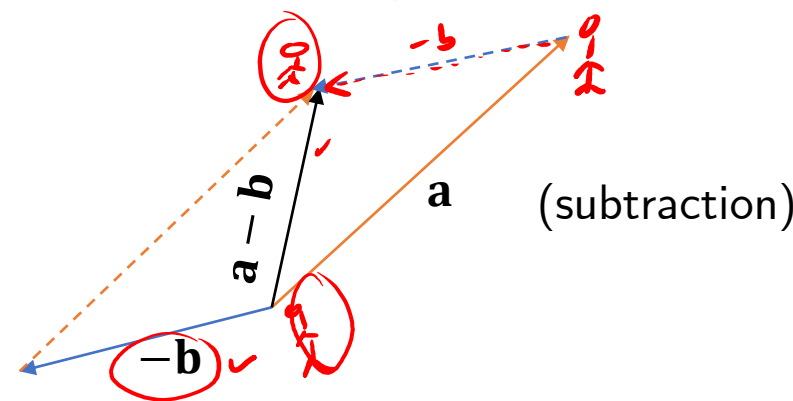
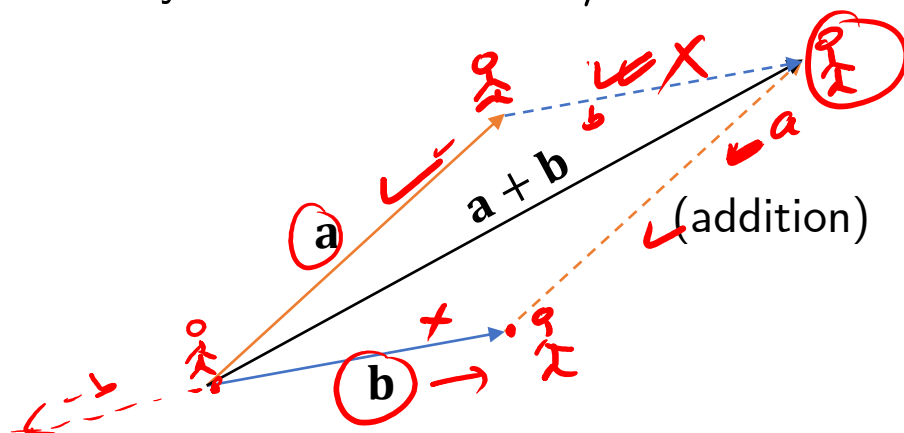
Vector Algebra



- **Scaling** (Scalar Multiplication : $\alpha \mathbf{v}$) : Multiplying the vector by a constant scales its length only. However, the direction will flip if you multiply a negative scalar.

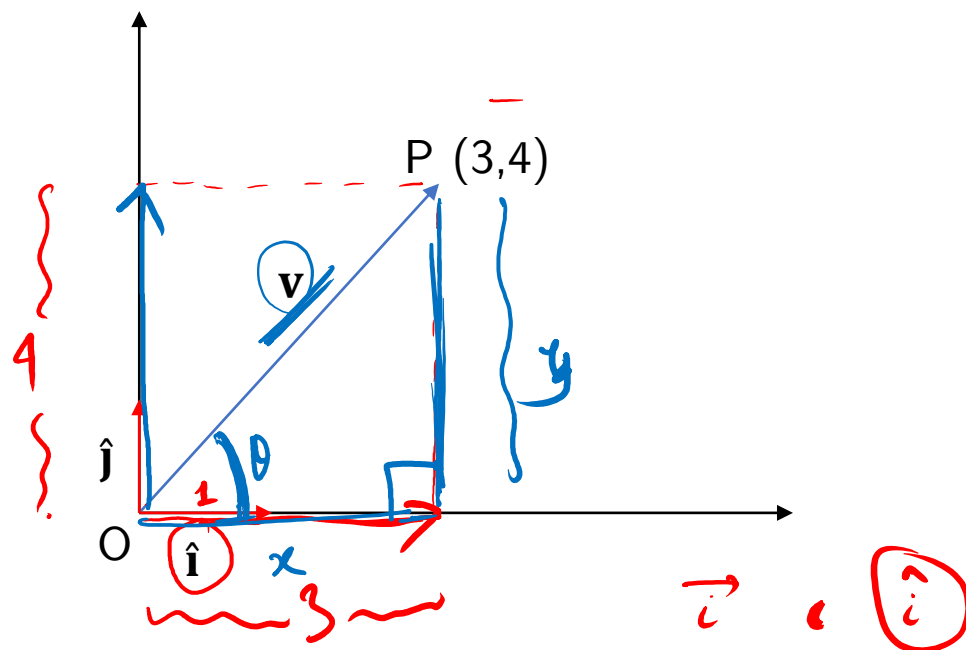


- **Addition / Subtraction** : We can also add and subtract vectors i.e., $\mathbf{a} + \mathbf{b}$ or $\mathbf{a} - \mathbf{b}$. Geometrically, Vector additions/subtractions can be realized using the notion of displacement.



Vector Algebra

- Mathematically, vectors can be written in several ways. Some of the conventions we typically follow is given below for vectors in 2D.



$$\underline{\underline{\mathbf{v} = (3, 4)}}$$

(row vector)

$$\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

(column vector)

$$\mathbf{v} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$$

(using basis vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ which are also called unit vectors along X and Y)

$$\mathbf{v} = 3 + 4i$$

(complex number representation. But complex number representation works only in 2D)

- Length or magnitude of the vector $|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5$

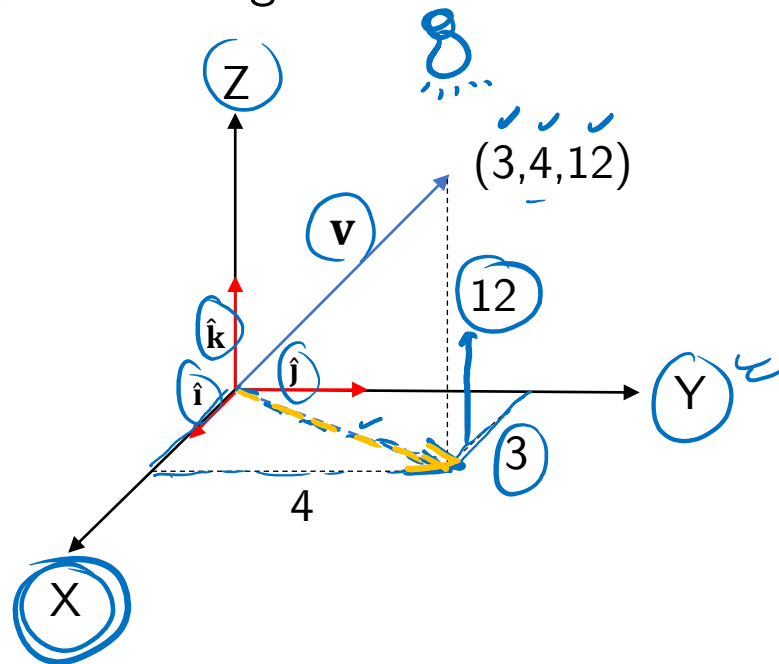
$$\cos\theta = \frac{x}{5}, \quad \sin\theta = \frac{y}{5}$$

$$x = 5\cos\theta, \quad y = 5\sin\theta$$

$$\cos\theta = \frac{3}{5}$$

Vector Algebra

- ▶ All the algebra extends to vectors in higher dimensions. Below is an example of Vectors in 3D



$$\mathbf{v} = \begin{pmatrix} 3 \\ 4 \\ 12 \end{pmatrix}$$

Pythagorean triplets
 3, 4, 5
 6, 8, 10
 5, 12, 13

$$\mathbf{v} = 3\hat{i} + 4\hat{j} + 12\hat{k}$$

shadow on xy plane

- ▶ The components of vector v along X, Y and Z are 3, 4 and 12.
- ▶ Length or magnitude of the vector $|\mathbf{v}| = \sqrt{3^2 + 4^2 + 12^2} = 13$
- ▶ Projection on XY plane ? 5

Vector Algebra

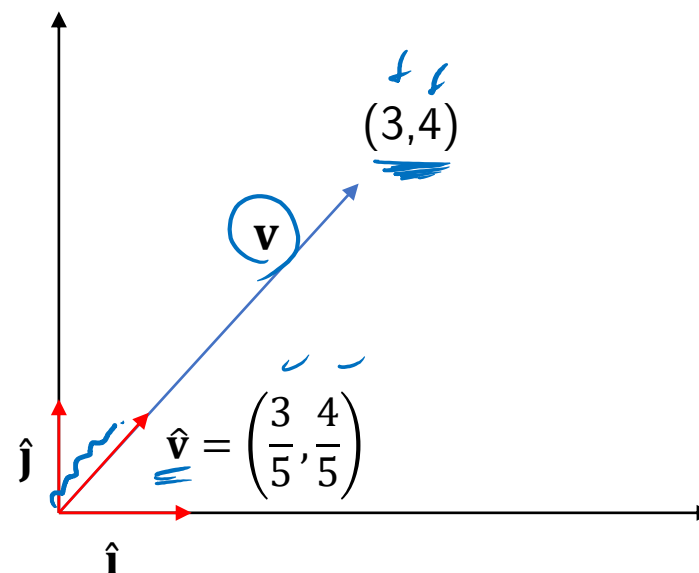
- A unit vector is a vector of length 1. The unit vectors along X and Y are given as $\hat{i}$ and $\hat{j}$. How do we find the unit vector along any vector? Well, divide the vector by its length.

unit vector along $\mathbf{v}$ is $\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}$

$$\hat{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \hat{j} \uparrow \rightarrow$$

$$\hat{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|\hat{\mathbf{v}}| = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = 1$$



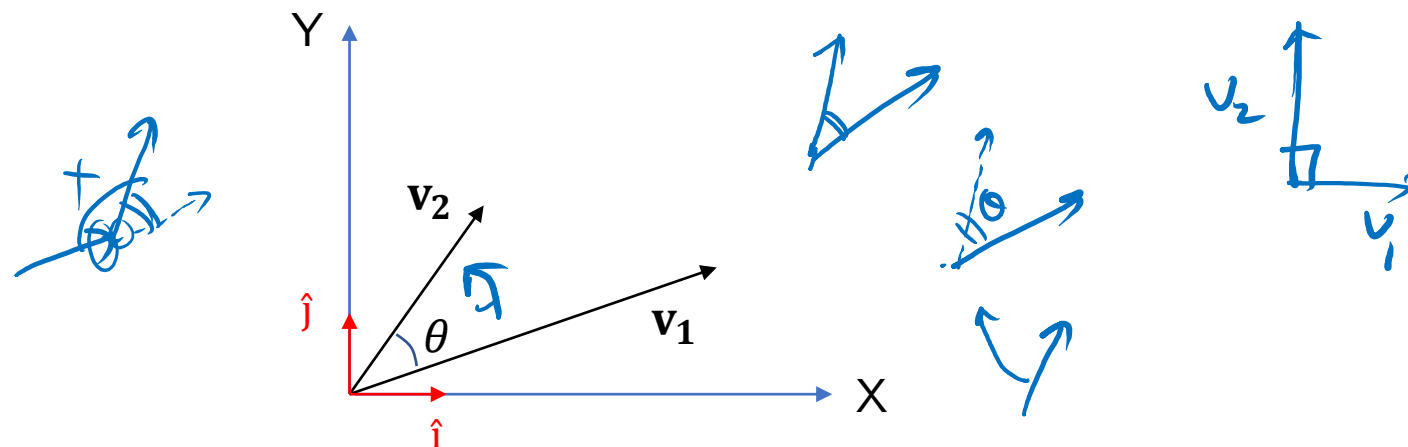
Exercise – Find the unit vector along $\underline{2\mathbf{a} + \mathbf{b}}$, where $\mathbf{a} = (1, 1)$, $\mathbf{b} = (4, 6)$

Vector Multiplication – Dot Product

- ▶ Vector dot product or inner product is denoted as **$\mathbf{a} \cdot \mathbf{b}$** . In Excel it's equal to SUMPRODUCT. The dot product is also called as scalar product, because the result is a scalar (number). For Example,

$$\text{if } \mathbf{v}_1 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \text{ and } \mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \text{ then } \mathbf{v}_1 \cdot \mathbf{v}_2 = 2 \times 1 + 2 \times 3 = 8$$

- ▶ Geometrically, the scalar product is related to the cosine of the angle between the vectors. It's given as $\mathbf{v}_1 \cdot \mathbf{v}_2 = |\mathbf{v}_1| |\mathbf{v}_2| \cos(\theta)$. Clearly, if two vectors are perpendicular then $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$



Vector Multiplication – Dot Product

Exercise – Find the following dot products if $\mathbf{v}_1 = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix}$, $|\mathbf{v}_1| = 2$, $|\mathbf{v}_2| = \sqrt{4+12} = 4$

a) $\mathbf{v}_1 \cdot \mathbf{v}_1 = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} = 3 + 1 = 4$, $\mathbf{v}_1 \cdot \mathbf{v}_1 = |\mathbf{v}_1| |\mathbf{v}_1| \cos(0^\circ) = |\mathbf{v}_1|^2 = 3 + 1 = 4$

b) $\widehat{\mathbf{v}}_1 \cdot \widehat{\mathbf{v}}_2 = \frac{\begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix}}{|\begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix}|} = \frac{4\sqrt{3}}{4} = \sqrt{3}/2$

c) $\mathbf{v}_1 \cdot \hat{\mathbf{i}} = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \sqrt{3}$ $\rightarrow |\mathbf{v}_1| |\hat{\mathbf{i}}| \cos(0) = \sqrt{3}$

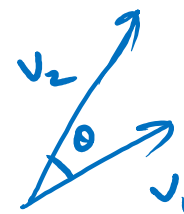
d) $\mathbf{v}_1 \cdot \hat{\mathbf{j}} = 1$

e) $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0 + 0 = 0$
 $|\hat{\mathbf{i}}| |\hat{\mathbf{j}}| \cos 90^\circ = 0$

f) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$

g) $\hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = 1$



Exercise – Find the angle between $\underline{(1,1,2)}$ and $\underline{(2,-1,1)}$

Span of Vectors

- The span of vector(s) is the space that can be accessed by a linear combinations of those. In other words, if $\mathbf{v}_1$ and $\mathbf{v}_2$ are two vectors then the span of $(\mathbf{v}_1, \mathbf{v}_2)$ is all vectors that can be created by $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2$. For example,

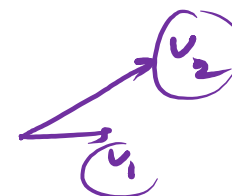


Span of $\hat{\mathbf{i}}$ is X-axis and span of $\hat{\mathbf{j}}$ is Y-axis

Span of $(\hat{\mathbf{i}}, \hat{\mathbf{j}})$ is entire XY plane or $\mathbb{R}^2$

$$5\hat{\mathbf{i}} + 17\hat{\mathbf{j}}$$

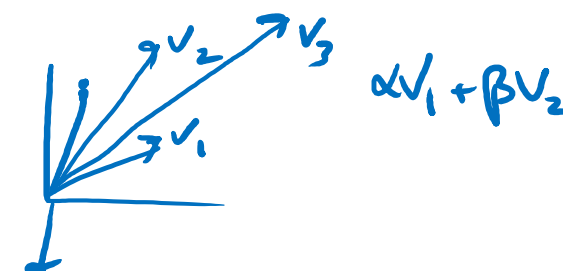
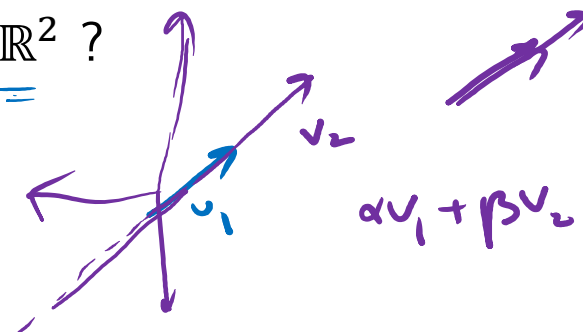
$$(5, 17)$$



Exercise – Express $\mathbf{v}_3$ as a linear combination of $(\mathbf{v}_1, \mathbf{v}_2)$, where $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} 7 \\ 13 \end{pmatrix}$

- Is span of any 2 vectors in 2D always $\mathbb{R}^2$?

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2\mathbf{v}_1$$



Independence of vectors

- A set of vectors are linearly independent if none of the vectors can be represented as a linear combination of other vectors. e.g., if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are 3 vectors and if $\mathbf{c} \neq \alpha\mathbf{a} + \beta\mathbf{b}$ for any (α, β) , then $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are linearly independent.

$$(\hat{i}, \hat{j}, \hat{k}) \propto \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Exercise – Which of the following set of vectors are linearly independent ?

a) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ → dependent (shown by a diagram where the vectors are in a plane and one is a linear combination of the others)

b) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ → independent

c) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ → dependent

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

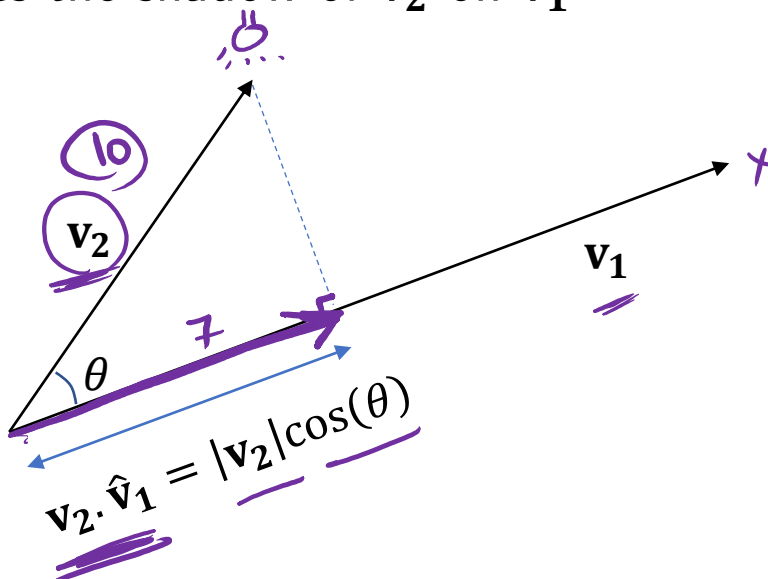
$$= 4 \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$\begin{aligned} 2\alpha + \beta &= 3 \\ 2\alpha + 4\beta &= 4 \\ 2\beta &= -2, \beta = -1 \\ \alpha &= 4 \end{aligned}$$

d) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ →

Component and Projection

- ▶ The component or scalar projection of a vector $\mathbf{v}_2$ along a vector $\mathbf{v}_1$ is given by $\mathbf{v}_2 \cdot \hat{\mathbf{v}}_1$. Note that it's a scalar and basically represents the shadow of $\mathbf{v}_2$ on $\mathbf{v}_1$

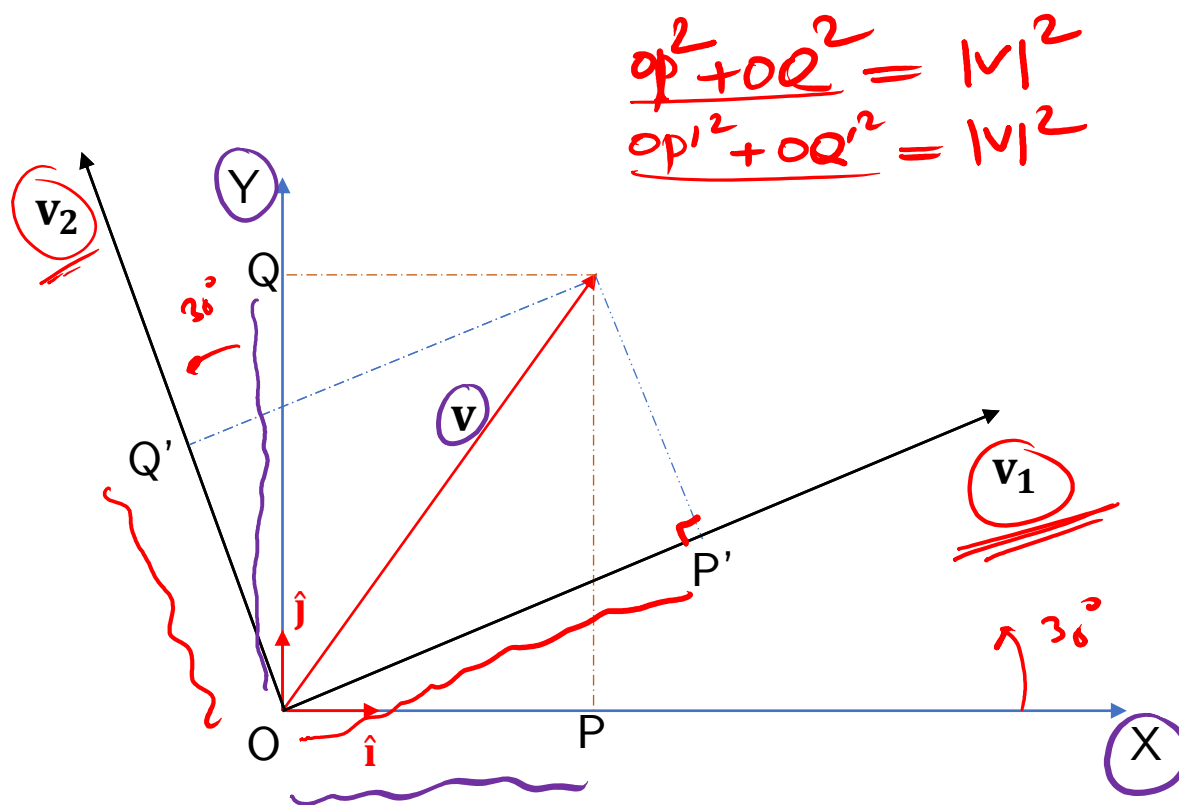
$$\begin{aligned}
 & \text{Handwritten: } \mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_2 \cdot \mathbf{v}_1 \\
 & \text{Handwritten: } \underline{|\mathbf{v}_2| \cos \theta} \\
 & = |\mathbf{v}_2| \cdot \left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{|\mathbf{v}_1| |\mathbf{v}_2|} \right) \\
 & = \mathbf{v}_2 \cdot \left(\frac{\mathbf{v}_1}{|\mathbf{v}_1|} \right) = \mathbf{v}_2 \cdot \hat{\mathbf{v}}_1
 \end{aligned}$$


- ▶ The vector projection is obtained by including the direction of $\mathbf{v}_1$ i.e., $(\mathbf{v}_2 \cdot \hat{\mathbf{v}}_1) \hat{\mathbf{v}}_1$ scalar.

Exercise – Find the projection of $\mathbf{a} = (1, 2)$ on $\mathbf{b} = (3, 4)$

Change of bases

- X and Y axes are the default bases. We can use any other orthogonal pair of axes as basis

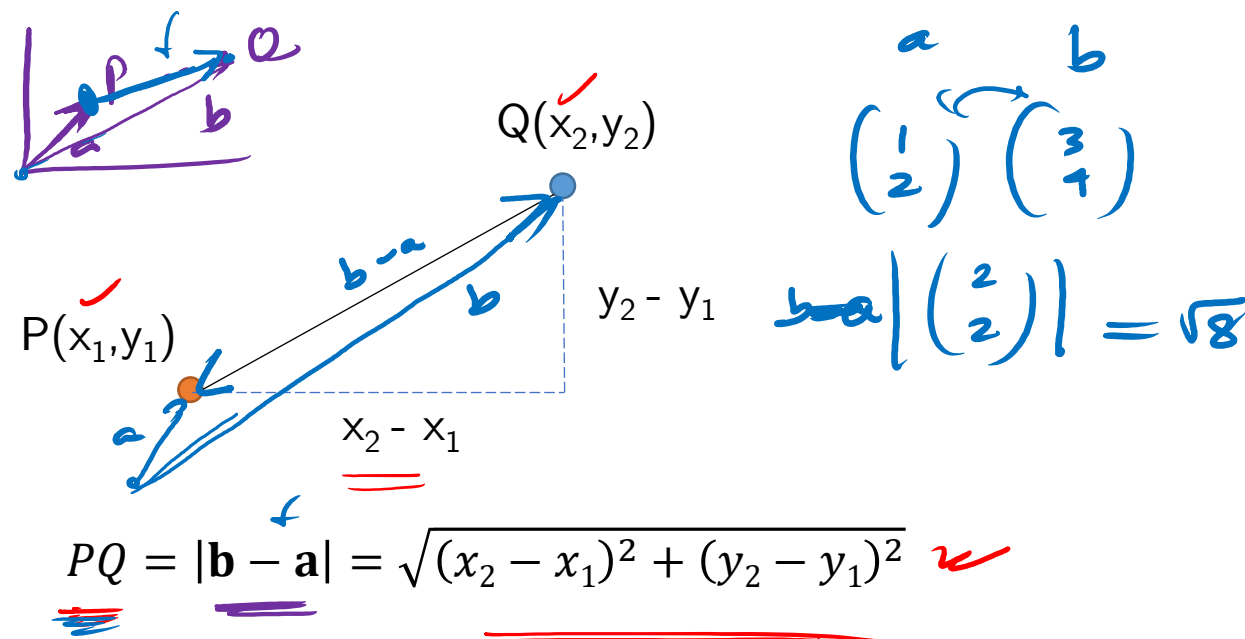
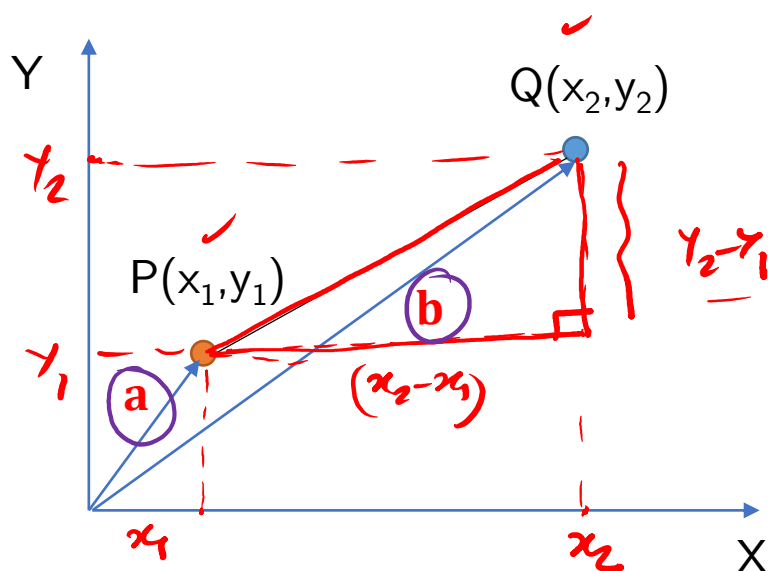


$$\underline{v} = \underline{OP} \hat{i} + \underline{OQ} \hat{j} \quad (\text{default basis})$$

$$\underline{v} = \underline{OP'} \hat{v}_1 + \underline{OQ'} \hat{v}_2 \quad (\text{new basis})$$

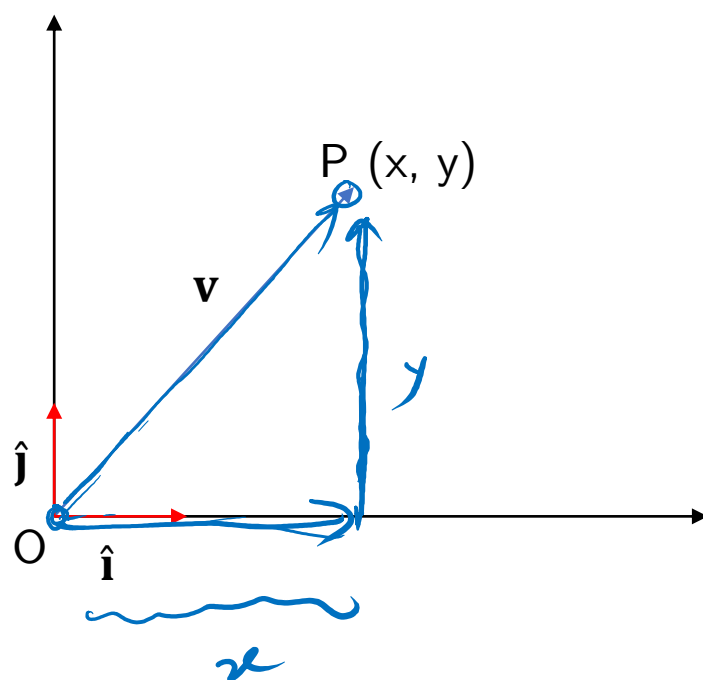
Distance

► The distance between 2 point vectors **a**, **b** is given as $|\mathbf{b} - \mathbf{a}|$



Vector Norm

► Vector norms are defined as follows



$$\ell^1 \text{ norm} := ||v||_1 = (|x| + |y|)^{1/1} \quad (\text{Manhattan norm}) \quad \text{taxi distance}$$

$$\ell^2 \text{ norm} := ||v||_2 = (|x|^2 + |y|^2)^{1/2} \quad (\text{Euclidean norm})$$

$$\ell^p \text{ norm} := ||v||_p = (|x|^p + |y|^p)^{1/p}$$

$$\ell^\infty \text{ norm} := ||v||_\infty = \max(|x|, |y|)$$

$$(|x|^4 + |y|^4)^{1/4} \rightarrow \max(|x|, |y|)$$

$$(4^{10000} + 5^{10000})^{1/10000} = 5$$

Concept Checker

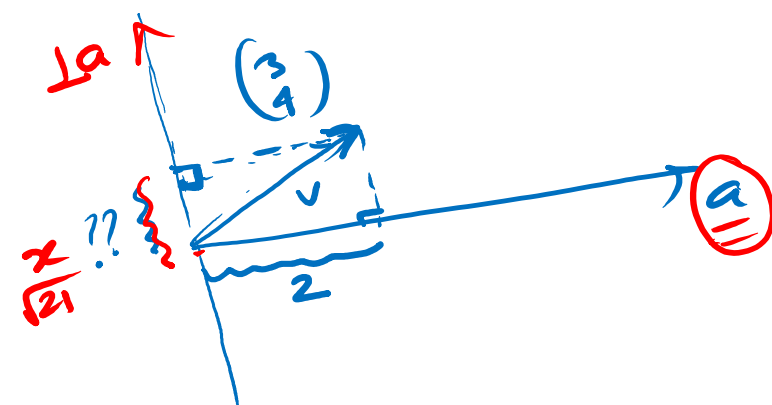


Exercises

1. Find the unit vector perpendicular to $\mathbf{v} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ in XY plane
2. Find the new vector when $\mathbf{v} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ is rotated counterclockwise about origin by 30°
3. Find the value of b if the following vectors are orthogonal $\mathbf{v}_1 = \begin{pmatrix} b \\ -1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$
4. The scalar projection of vector $\mathbf{v} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ on $\mathbf{a}$ is equal to 2. What is the scalar projection of $\mathbf{v}$ on a vector perpendicular to $\mathbf{a}$?

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$$

$$3b - 6 = 0, b = 2$$



$$3^2 + 4^2 = 2^2 + x^2$$

$$\Rightarrow x^2 = 21$$

$$x = \sqrt{21}$$